



Node.js Tutorial

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Node.js is an open-source, cross-platform JavaScript runtime built on Chrome's V8 engine. It enables developers to run JavaScript outside the browser to build fast, scalable server-side applications.

- JavaScript was initially frontend-only language, and Node.js (2009) enabled backend development as well.
- Non-blocking, event-driven architecture for high performance.
- Supports the creation of [REST APIs](#), real-time applications, and [microservices](#).
- Comes with a rich library of modules through npm (Node Package Manager).



First code example:

```
// Import the http module
const http = require("http");

// Create a server
const server = http.createServer((req, res) => {
  res.statusCode = 200;
  res.setHeader("Content-Type", "text/plain");
  res.end("Welcome to the Node.js Tutorial");
});

// Listen on port 3000
```



```
server.listen(3000, () => {  
  console.log(  
    "Server is running on http://localhost:3000");  
});
```

It will start a server, and when you visit <http://localhost:3000>, it will display

Welcome to the Node.js Tutorial

- The http module is imported to create a basic HTTP server.
- The createServer() method is used to handle incoming requests and send responses.
- The server listens on port 3000, and a message is displayed in the browser when accessed.

Need to Learn Node.js

It allows developers to build fast, scalable server-side applications using JavaScript.

- Enables full-stack development with a single language.
- Ideal for real-time apps like chat and gaming servers.
- Handles I/O-intensive tasks efficiently with non-blocking architecture.
- Supported by a large community and vast npm ecosystem.

Introduction to Node.js

Introduction to Node.js provides an overview of its runtime environment, architecture, and how it enables server-side JavaScript development.

- [Node.js Introduction](#)
- [Importance of Node.js](#)
- [Working](#)

Getting Started

Understand the basic concepts of Node.js and begin by creating your first simple application.

- [Basics](#)
- [First Application](#)
- [REPL](#)

Node Package Manager

Manage project dependencies and packages efficiently using NPM.

- [NPM \(Node.js Package Manager\)](#)
- [Create and Publish NPM packages](#)
- [Global Installation of Dependencies](#)

Modules in Node.js

Understand how to organize and reuse code using different types of modules.

- [Modules](#)
- [Core modules](#)
- [Third Party modules](#)
- [Custom modules](#)

Asynchronous Programming

Learn how Node.js handles multiple tasks efficiently using non-blocking operations.

- [Blocking and Non-Blocking](#)
- [Callback Concept](#)
- [Promise Chaining](#)
- [Event Loop](#)

Core Concept

Understand essential Node.js concepts that control scope, context, and global behavior.

- [This Binding](#)
- [Global Objects](#)
- [Session Variable](#)

Building Applications

Create and manage server-side applications using Node.js features and tools.

- [Web Server](#)
- [Frameworks](#)
- [Child Process](#)

Express.js

Build web applications efficiently using the Express.js framework.

- [Introduction to Express.js](#)
- [MVC Design Pattern](#)
- [Serving Static Files](#)
- [Middleware](#)
- [EJS Template](#)

Debugging and Utilities

Tools and techniques to debug code and improve development workflow.

- [Debugging](#)
- [Set Console Font Color](#)
- [Automatic Restart Server with Nodemon](#)

Node.js Complete References

It covers the core modules and built-in APIs used for server-side development.

System & Core Modules

Handle core functionalities related to system operations and runtime environment.

- [Assert](#)
- [Console](#)
- [Utility](#)
- [OS](#)
- [Process](#)
- [V8](#)
- [VM Module](#)

File & Data Handling

Manage data processing, file operations, and data transformation.

- [Buffer](#)
- [String Decoder](#)
- [Query String](#)
- [File System](#)
- [Path Module](#)
- [Stream](#)
- [URL](#)

Networking, Security & Performance

Support communication, security features, and performance optimization.

- [HTTP Module](#)
- [HTTP2](#)
- [DNS](#)
- [UDP/DataGram](#)
- [TLS/SSL Module](#)
- [Crypto](#)

- [Zlib](#)
- [Timers Module](#)

Node.js Interview Questions

- [Interview Questions and Answers](#)

Node.js Online Quiz Questions

- [Quiz | Set-1](#)
- [Quiz | Set-2](#)
- [Quiz | Set-3](#)

Node.js Projects

- [Library Management System](#)
- [User Management System](#)
- [Stock Market Portfolio App](#)
- [Chat App using socket.io](#)
- [Mail sender](#)
- [Task Manager](#)
- [Video Call App](#)
- [AI Whatsapp Bot](#)
- [Real Time News Aggregator](#)

Node.js Advantages

- **Easy Scalability:** It compiles and executes JavaScript at lightning speeds, making it highly scalable.
- **Real-time Web Apps:** It enables real-time communication for chat, gaming, social media updates, and more.
- **Microservices:** It is lightweight and ideal for microservice architectures.
- **JavaScript Everywhere:** Learn JavaScript once, and you can use it both for front-end and back-end development.
- **Efficient Data Streaming:** It efficiently handles I/O processes like media transcoding during uploads.
- **Event-Driven Architecture:** Unlike traditional servers, Node.js handles concurrent requests effectively.
- **Strong Community Support:** It has an independent community backing its development.

Prerequisites

- [JavaScript](#)

- [Basic Web development](#)

Installation

To start with Node.js, you need to install and set it up on your machine.

- Installation on [Windows](#), [Linux](#), and [MacOS](#)

Important Links

- [Careers and Jobs in Node.js](#)
- [Companies That Use Node.js](#)
- [Node.js Vs Django Vs Spring](#)

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